Darshan UNIVERSITY

योग: कर्मसु कौशलम्

Python Programming - 2301CS404

Lab - 7

Roll No : 324

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01) WAP to find sum of tuple elements.

Sample Input:

```
T = (10, 20, 30, 40)
```

Sample Output:

```
Sum of tuple elements = 100
```

In [2]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))
ans=0
for i in t1:
    ans = ans + i
print('Sum of tuple elements',ans)
```

```
Sum of tuple elements 100
```

02) WAP to find Maximum and Minimum K elements in a given tuple.

Sample Input:

```
T = (7, 2, 9, 4, 1, 5)
K = 2
```

Sample Output:

```
Minimum 2 elements: (1, 2)
Maximum 2 elements: (9, 7)
```

In [6]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

max1 = 0
max2 = 0
min1 = max(t1)
min2 = max(t1)

for i in t1:
    if max2<i:
        max1 = max2
        max2 = i
    if min2>i:
        min1 = min2
        min2 = i
ans1 = (min2,min1)
ans2 = (max2,max1)
print(ans1)
print(ans2)
```

```
(1, 2)
(9, 7)
```

03) WAP to find tuples which have all elements divisible by K from a list of tuples.

Sample Input:

```
tuple_list = [(2, 4, 6), (3, 6, 9), (4, 8, 12), (5, 10, 15)]
K = 2
```

Sample Output:

```
Tuples divisible by 2 : [(2, 4, 6), (4, 8, 12)]
```

In [8]:
```
s = int(input("Enter Size:"))
k = int(input("Enter K:"))
l1 = list()
for i in range(s):
    t1 = tuple(map(int,input("Enter Data using space:").split()))
    l1.append(t1)
ans = list()
for i in l1:
    for j in i:
        if j%k!=0:
```

```
            break
    else:
        ans.append(i)
print(ans)
```

```
[(2, 4, 6), (4, 8, 12)]
```

04) WAP to create a list of tuples from given list having number and its cube in each tuple.

Sample Input:

```
numbers = [1, 2, 3, 4, 5]
```

Sample Output:

```
List of tuples with number and its cube: [(1, 1), (2, 8), (3, 27),
(4, 64), (5, 125)]
```

In [10]:
```
l1 = list(map(int,input("Enter Data using space:").split()))

ans = list()

for i in l1:
    t1 = (i,i**3)
    ans.append(t1)
print(ans)
```

```
[(1, 1), (2, 8), (3, 27), (4, 64), (5, 125)]
```

05) WAP to find tuples with all positive elements from the given list of tuples.

Sample Input:

```
tuple_list = [(1, 2, 3), (-1, 2, 3), (4, 5, 6), (0, 1, 2), (-3, -4,
-5)]
```

Sample Output:

```
Tuples with all positive elements: [(1, 2, 3), (4, 5, 6)]
```

In [12]:
```
s = int(input("Enter Size:"))
l1 = list()
for i in range(s):
    t1 = tuple(map(int,input("Enter Data using space:").split()))
    l1.append(t1)
ans = list()
for i in l1:
```

```
    for j in i:
        if j<0 or j==0:
            break
    else:
        ans.append(i)
print(ans)
```

```
[(1, 2, 3), (4, 5, 6)]
```

06) WAP to remove tuples of length K.

Sample Input:

```
tuple_list = [(1, 2), (3, 4, 5), (6,), (7, 8, 9), (10, 11)]
K = 2
```

Sample Output:

```
List after removing tuples of length 2 : [(3, 4, 5), (6,), (7, 8,
9)]
```

In [14]:
```
s = int(input("Enter Size:"))
k = int(input("Enter K:"))
l1 = list()
for i in range(s):
    t1 = tuple(map(int,input("Enter Data using space:").split()))
    l1.append(t1)

ans= list()
for i in l1:
    if len(i)!=k:
        ans.append(i)
print(ans)
```

```
[(3, 4, 5), (6,)]
```

07) WAP to remove all occurrences of a given element from a tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
x = 3
```

Sample Output:

```
(5, 7, 9)
```

In [15]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

x = int(input("Enter X:"))
```

```
ans = list()

for i in t1:
    if i!=x:
        ans.append(i)
print(tuple(ans))
```

```
(5, 7, 9)
```

08) WAP to remove duplicates from tuple.

Sample Input:

```
T = (3, 5, 3, 7, 3, 9)
```

Sample Output:

```
(3, 5, 7, 9)
```

In [16]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

ans = list()

for i in t1:
    if i not in ans:
        ans.append(i)
print(tuple(ans))
```

```
(3, 5, 7, 9)
```

09) WAP to multiply adjacent elements of a tuple and print that resultant tuple.

Sample Input:

```
t = (2, 3, 4, 5)
```

Sample Output:

```
Resultant tuple after multiplying adjacent elements: (6, 12, 20)
```

In [18]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

ans = list()

for i in range(1,len(t1)):
    ans.append(t1[i-1]*t1[i])
print(tuple(ans))
```

```
(6, 12, 20)
```

10) WAP to test if the given tuple is distinct or not.

Sample Input:

```
tuple1 = (1, 2, 3, 4)
tuple2 = (1, 2, 2, 3)
```

Sample Output:

```
(1, 2, 3, 4) is distinct? -> True
(1, 2, 2, 3) is distinct? -> False
```

In [20]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

for i in t1:
    if t1.count(i)>1:
        print("Not Distinct")
        break;
else:
    print("Distinct")
```

```
Not Distinct
```

11) WAP to rotate the elements of a tuple to the right by `k` positions.

Sample Input:

```
T = (10, 20, 30, 40, 50)
k = 2
```

Sample Output:

```
(40, 50, 10, 20, 30)
```

In [25]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))

k = int(input("Enter K:"))

ans = list()

ans = t1[len(t1)-k:] + t1[0:len(t1)-k]

print(ans)
```

```
(40, 50, 10, 20, 30)
```

12) WAP to merge two tuples of equal length alternately.

Sample Input:

```
T1 = (1, 3, 5)
T2 = (2, 4, 6)
```

Sample Output:

```
(1, 2, 3, 4, 5, 6)
```

In [28]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))
t2 = tuple(map(int,input("Enter Data using space:").split()))
i = 0
j = 0
flag = 0
ans = list()
for k in range(len(t1)*2):
    if flag==0:
        ans.append(t1[i])
        i = i + 1
        flag = 1
    else:
        ans.append(t2[j])
        j = j + 1
        flag = 0
print(tuple(ans))
```

```
(1, 2, 3, 4, 5, 6)
```

13) WAP to create a tuple of squares of elements that are even and greater than 10.

Sample Input:

```
T = (4, 12, 7, 18, 10)
```

Sample Output:

```
(144, 324)
```

In [29]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))
ans = list()
for i in t1:
    if i>10 and i%2==0:
        ans.append(i**2)
print(tuple(ans))
```

```
(144, 324)
```

14) WAP to create (index, value) pairs for prime numbers.

Sample Input:

```
T = (4, 7, 9, 11, 15)
```

Sample Output:

```
((1, 7), (3, 11))
```

In [33]:
```
t1 = tuple(map(int,input("Enter Data using space:").split()))
ans = list()

for i in range(len(t1)):
    for j in range(2,t1[i]):
        if t1[i]%j==0:
            break
    else:
        t2 = (i,t1[i])
        ans.append(t2)
print(tuple(ans))
```

```
((1, 7), (3, 11))
```

15) WAP to split a tuple into even-index and odd-index tuples.

Sample Input:

```
T = ('a', 'b', 'c', 'd', 'e')
```

Sample Input:

```
even = ('a', 'c', 'e')
odd = ('b', 'd')
```

In [35]:
```
t1 = tuple(input("Enter Data using space:").split())
odd = list()
even = list()
for i in range(len(t1)):
    if i%2!=0:
        odd.append(t1[i])
    else:
        even.append(t1[i])
print("Even = ",tuple(even))
print("Odd = ",tuple(odd))
```

```
Even =  ('a', 'c', 'e')
Odd =  ('b', 'd')
```

16) WAP to compute column-wise sum from a tuple of lists.

Sample Input:

```
T = ([1, 2, 3], [4, 5, 6], [7, 8, 9])
```

Sample Output:

```
12 15 18
```

In [44]:
```
s = int(input("Size:"))
outer = list()
for i in range(s):
    inner = list(map(int,input("Enter Data using space:").split()))
    outer.append(inner)

ans = [0]*len(outer[0])
for i in range(len(outer)):
    for j in range(len(outer[i])):
        ans[j] = ans[j] + outer[i][j]
print(ans)
```

```
[5, 7, 9]
```

In [ ]: